EEE4775 Final Supporting Work

Section	Topic
System Architecture	Layout of System Design
Latency Measurements	WITH_LOAD = 0 & WITH_LOAD = 1
Induced Failure	Removal of ISR Yield Latency Measurements
Concurrency Diagram	Rate Monotonic Scheduling Demonstration
VCD File Photos	Interrupt Response Time

1. System Architecture

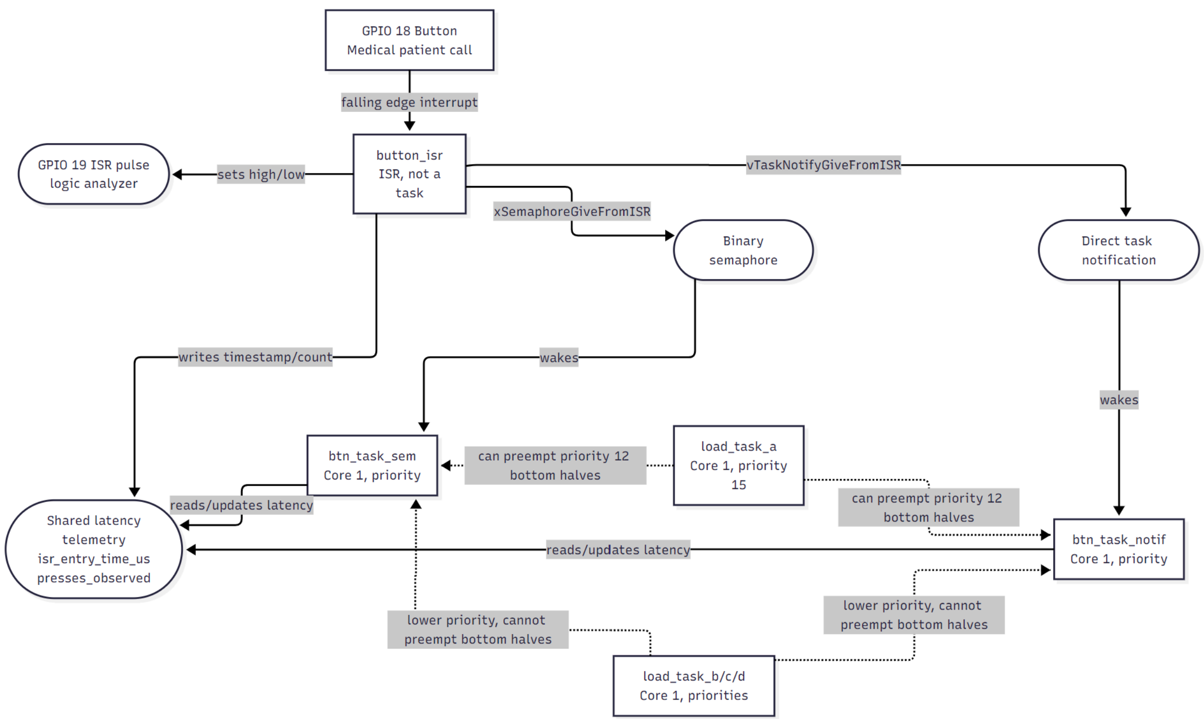

Fig. 1 System Diagram

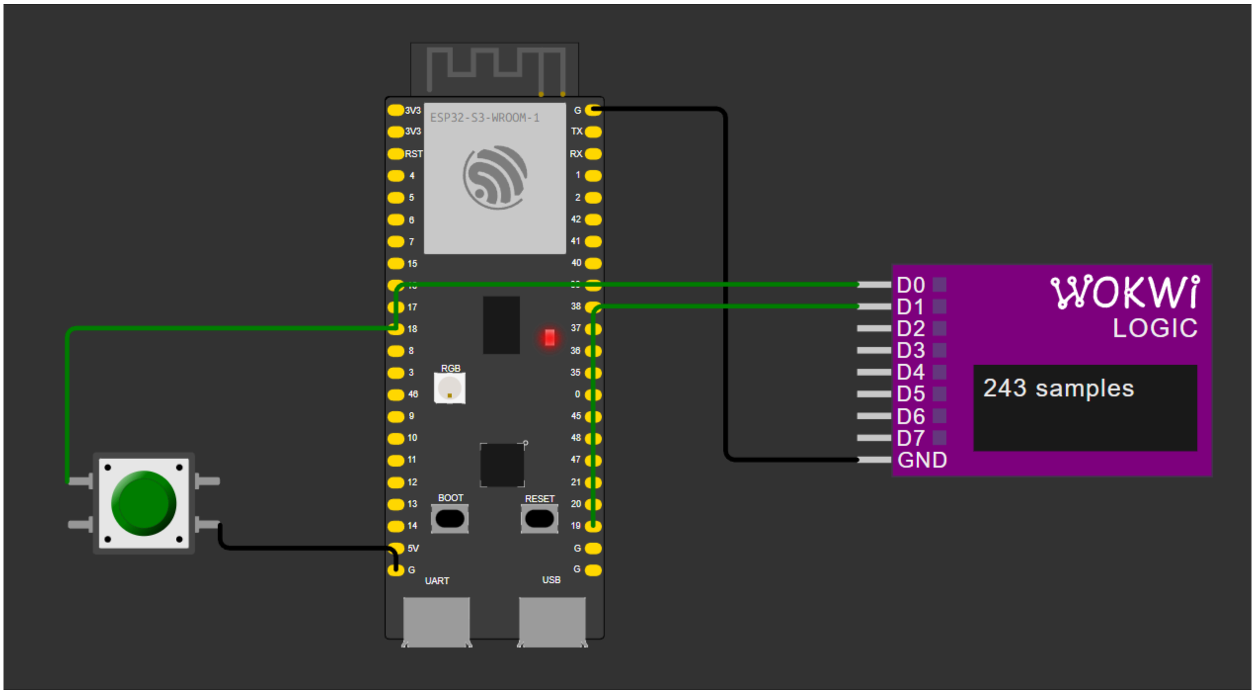

Fig. 2 Simulated Device

2. Latency Measurements

2.1 WITH_LOAD = 0 & INJECT_MISSING_ISR_YIELD = 0

```
I (20) final: ==== [Medical Alert] starting  ISR + bottom-half ====
I (20) final: Run mode: IDLE (WITH_LOAD=0)  baseline latency, no background tasks
I (30) final: Fault injection Disabled: ISR exit yield present
I (30) final: Press the button on GPIO 18. Scope GPIO 19 to time the ISR.
```

Fig. 3 Startup Message with No Load

```
I (118150) final: [notif] patient call button press #95  latency=30 us (max=30) (notif count=1)
I (118150) final: [sem] patient call button press #95  latency=2574 us (max=2574)
I (118590) final: [notif] patient call button press #96  latency=30 us (max=30) (notif count=1)
I (118600) final: [sem] patient call button press #96  latency=2484 us (max=2574)
I (119020) final: [notif] patient call button press #97  latency=30 us (max=30) (notif count=1)
I (119020) final: [sem] patient call button press #97  latency=632 us (max=2574)
I (119440) final: [notif] patient call button press #98  latency=25 us (max=30) (notif count=1)
I (119450) final: [sem] patient call button press #98  latency=2395 us (max=2574)
I (119840) final: [notif] patient call button press #99  latency=30 us (max=30) (notif count=1)
I (119840) final: [sem] patient call button press #99  latency=2574 us (max=2574)
I (120810) final: [notif] patient call button press #100  latency=30 us (max=30) (notif count=1)
I (120810) final: [sem] patient call button press #100  latency=2604 us (max=2604)
```

Fig. 4 No Load Latency after 100 Presses

2.2 WITH_LOAD = 1 & INJECT_MISSING_ISR_YIELD = 0

```
I (20) final: ==== [Medical Alert] starting  ISR + bottom-half ====
I (20) final: Run mode: UNDER LOAD (WITH_LOAD=1)  App 2's 4 tasks on Core 1
I (30) final: Fault injection Disabled: ISR exit yield present
```

Fig. 5 Startup Message with Load

```
I (85030) final: [notif] patient call button press #99  latency=37 us (max=2450) (notif count=1)
I (85030) final: [sem] patient call button press #99  latency=132 us (max=2646)
I (85450) final: [notif] patient call button press #100  latency=30 us (max=2450) (notif count=1)
I (85450) final: [sem] patient call button press #100  latency=503 us (max=2646)

=== Medical Alert System: Background Task Monitor ===
Task  Period   Priority  Heartbeats   WCET(us)
A     10 ms    15        8600         138
B     20 ms    10        4300         12599
C     50 ms    5         1720         14548
D     100 ms   2         860          10413
(heartbeats should grow monotonically; a stalled counter = starved or hung task)

Semaphore max latency: 2646 us
Notification max latency: 2450 us
```

Fig. 6 Load Latency after 100 Presses with Task Monitor

3. Induced Failure

3.1 No Load, ISR Yield Removed

```
I (20) final: ==== [Medical Alert] starting  ISR + bottom-half ====
I (20) final: Run mode: IDLE (WITH_LOAD=0)  baseline latency, no background tasks
W (30) final: Fault injection enabled: ISR exit yield removed
I (30) final: Press the button on GPIO 18. Scope GPIO 19 to time the ISR.
```

Fig. 7 Startup Message with No Load & ISR Yield Removed

```
I (55490) final: [notif] patient call button press #95  latency=31 us (max=2279) (notif count=1)
I (55490) final: [sem] patient call button press #95  latency=2605 us (max=2605)
I (56800) final: [notif] patient call button press #96  latency=30 us (max=2279) (notif count=1)
I (56800) final: [sem] patient call button press #96  latency=2604 us (max=2605)
I (57370) final: [notif] patient call button press #97  latency=30 us (max=2279) (notif count=1)
I (57370) final: [sem] patient call button press #97  latency=1265 us (max=2605)
I (57720) final: [notif] patient call button press #98  latency=30 us (max=2279) (notif count=1)
I (57720) final: [sem] patient call button press #98  latency=1260 us (max=2605)
I (57920) final: [notif] patient call button press #99  latency=30 us (max=2279) (notif count=1)
I (57920) final: [sem] patient call button press #99  latency=1257 us (max=2605)
I (60210) final: [notif] patient call button press #100  latency=24 us (max=2279) (notif count=1)
I (60210) final: [sem] patient call button press #100  latency=2629 us (max=2629)
```

Fig. 8 No Load & ISR Yield Removed Latency after 100 Presses

3.2 Under Load, ISR Yield Removed

```
I (20) final: ==== [Medical Alert] starting  ISR + bottom-half ====
I (20) final: Run mode: UNDER LOAD (WITH_LOAD=1)  App 2's 4 tasks on Core 1
W (30) final: Fault injection enabled: ISR exit yield removed
```

Fig. 9 Startup Message with Load & ISR Yield Removed

```
I (80710) final: [notif] patient call button press #100  latency=30 us (max=2467) (notif count=1)
I (80710) final: [sem] patient call button press #100  latency=2635 us (max=2728)

=== Medical Alert System: Background Task Monitor ===
Task  Period   Priority  Heartbeats   WCET(us)
A     10 ms    15        8100         138
B     20 ms    10        4050         12589
C     50 ms    5         1620         10595
D     100 ms   2         810          18737
(heartbeats should grow monotonically; a stalled counter = starved or hung task)

Semaphore max latency: 2728 us
Notification max latency: 2467 us
```

Fig. 10 With Load & ISR Yield Removed Latency after 100 Presses

4. Concurrency Diagram
4.1 Rate Monotonic Scheduling Demonstration

Figure 11 is a conceptual rate monotonic scheduling illustration and is not drawn to exact scale. The measured task duration may include preemption time and therefore are not used as isolated execution times in the diagram.

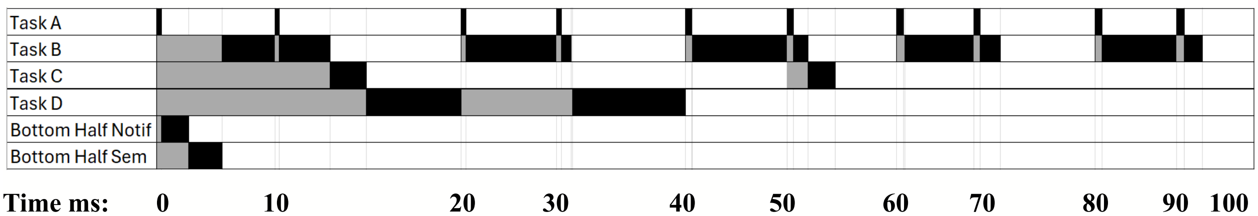

Fig. 11 Rate Monotonic Scheduling

Legend:

Black = Running

Gray = Ready

White = Blocked

4.2 Scheduling Pattern

Below is a description list of the pattern that can be seen in *Fig. 9* which is the rate monotonic scheduling occurring within the Medical Alert code when under load. Without load the diagram would not show tasks A-D and only consist of the two bottom half tasks occurring when there is a button press.

- Task A is running every 10 ms based on the period
- Task B is running every 20 ms based on the period and being preempted by A.
- Task C is running every 50 ms based on the period and being preempted by A and B.
- Task D is running every 100 ms (once in this frame) based on the period and being preempted by A, B, and C.
- The two bottom half tasks are both running once assuming only one button press has occurred and they are both only preempted by A.

5. VCD File Photos
5.1 Overview of Interrupt Latency Investigation

GPIO 18 represents the physical button input while GPIO 19 is driven high at the beginning of the ISR. The time between the GPIO 18 falling edge and GPIO 19 rising edge therefore represents interrupt response latency. One representative event was examined for both idle and loaded conditions.

5.2 Interrupt WITH_LOAD = 0

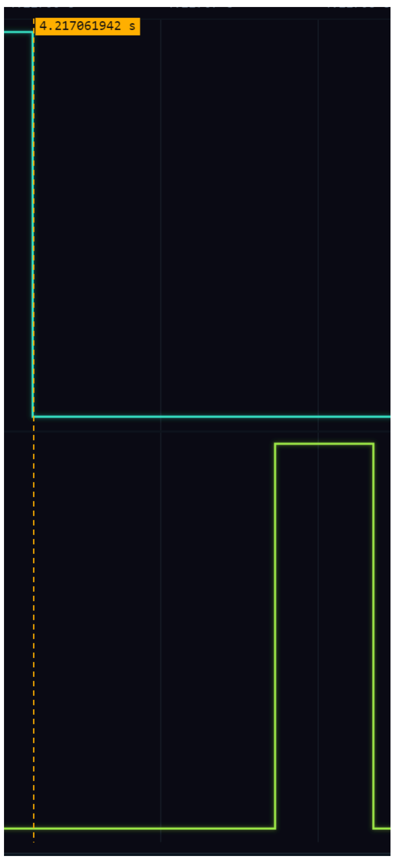

Fig. 12 Button Press Sensed

Fig. 13 Interrupt Starts

5.3 Interrupt WITH_LOAD = 1

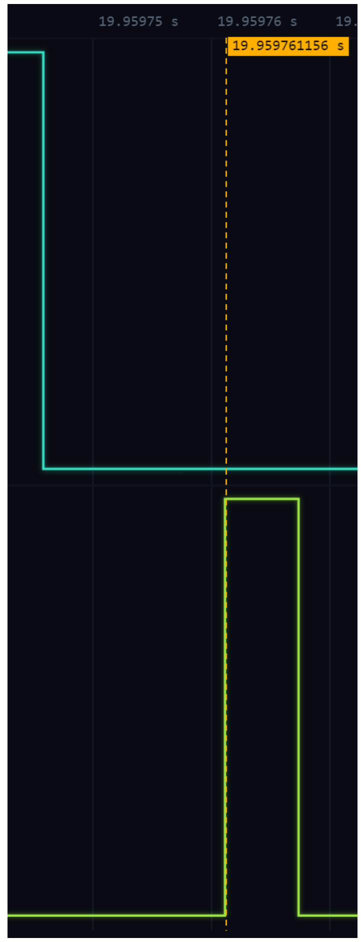

Fig. 14 Button Press Sensed *Fig. 15 Interrupt Starts*